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Control device, behavior guide system, and method for controlling same

Abstract

A control device includes a behavior recognition unit, a determination unit, a subsequent behavior annunciation unit, a correct behavior annunciation unit, and a warning unit. The behavior recognition unit recognizes a behavior executed by a shopper. The determination unit determines a subsequent behavior as the behavior for the shopper to subsequently perform based on the behavior recognized by the behavior recognition unit. The subsequent behavior annunciation unit makes an annunciation member disposed so as to correspond to a behavior area where the subsequent behavior is to be executed operate in a first annunciation configuration based on the subsequent behavior determined by the determination unit. The correct behavior annunciation unit makes the annunciation member operate in a second annunciation configuration when the behavior recognition unit recognizes that the shopper executes the subsequent behavior. The warning unit which makes the annunciation member disposed so as to correspond to the behavior area where a different behavior from the subsequent behavior is executed operate in a warning configuration when the behavior recognition unit recognizes that the shopper executes the different behavior.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2023-183466, filed on Oct. 25, 2023, the entire contents of which are incorporated herein by reference.

FIELD

(2) Embodiments of the present disclosure relate to a control device, a behavior guide system, and a method for controlling the same.

BACKGROUND

(3) In recent years, there are an increasing number of stores which are provided with self-service checkout terminals each called a self-service POS terminal for the user to execute the registration of an article to be purchased and settlement processing by him- or herself, and which adopt a sales configuration in which a shopper executes the registration of an article and the settlement processing. In general, in such a self-service POS terminal, the shoppers scan a code symbol attached to an article with a scanner to execute the registration of the article.

(4) In such a self-service POS terminal, a variety of techniques for preventing cheating are proposed. For example, there is proposed a technique of making a robot face follow the article gripped by a customer to thereby provide the user of the terminal with an impression of being monitored, and thus, achieving prevention of an occurrence of the cheating such as shoplifting. Further, there is also disclosed in this technique that the fact that some kind of operation failure or fraud is made is announced toward a sales clerk as an attendant or the like.

(5) In the past, there was no such a system that a shopper as a user was visually informed of whether the behavior was correct or wrong with respect to the operation at that time. Therefore, the prevention of an involuntary erroneous behavior and a willful illicit behavior of the user was not achieved.

Description

DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic configuration diagram showing a settlement system of a self-service type including a self-service POS terminal to which a behavior guide system according to a first embodiment including a control device according to the first embodiment is applied.

(2) FIG. 2 is a perspective view showing an example of an appearance configuration of the self-service POS terminal.

(3) FIG. 3 is a block diagram showing an example of circuit configuration of an essential part of the self-service POS terminal.

(4) FIG. 4 is a flowchart representing an example of a procedure of an essential part of information processing to be executed by a processor of the self-service POS terminal.

(5) FIG. 5 is a flowchart showing an example of a detailed procedure of registration behavior processing in FIG. 4.

(6) FIG. 6 is a schematic configuration diagram showing a settlement system of a self-service type including an information terminal to which a behavior guide system according to a second embodiment including a control device according to the second embodiment is applied.

(7) FIG. 7 is a perspective view showing an example of an appearance configuration of the information terminal attached to a shopping cart.

(8) FIG. 8 is a schematic diagram illustrating imaging ranges of a surrounding area camera and a basket area camera provided to the information terminal.

(9) FIG. 9 is a block diagram showing an example of a circuit configuration of an essential part of the information terminal.

(10) FIG. 10 is a flowchart representing an example of a procedure of an essential part of information processing to be executed by a processor of the information terminal.

(11) FIG. 11 is a flowchart showing an example of a detailed procedure of registration behavior processing in FIG. 10.

(12) FIG. 12 is a schematic configuration diagram showing a settlement system of a self-service type as a behavior guide system according to a third embodiment including a control device according to the third embodiment.

(13) FIG. 13 is a perspective view showing an example of an imaging range of a checkout

monitoring camera in FIG. 12.

(14) FIG. 14 is a sequence diagram illustrating an operation of the settlement system.

(15) FIG. 15 is a plan view showing an example of an appearance configuration of a hand-wash unit to which a behavior guide system according to a fourth embodiment including a control device according to the fourth embodiment is applied.

DETAILED DESCRIPTION

(16) A problem to be solved by the embodiments of the present disclosure is to provide a control device, a behavior guide system, and a method for controlling the behavior guide system which can prevent an involuntary erroneous behavior and a willful illicit behavior of a shopper as a user.

(17) In one embodiment, a control device is for controlling the annunciation members disposed so as to correspond respectively to a plurality of behavior areas where a plurality of behaviors included in a series of behaviors to be executed by a shopper is executed, and includes a behavior recognition unit, a determination unit, a subsequent behavior annunciation unit, a correct behavior annunciation unit, and a warning unit. The behavior recognition unit recognizes a behavior executed by a shopper. The determination unit determines a subsequent behavior as the behavior for the shopper to subsequently perform based on the behavior recognized by the behavior recognition unit. The subsequent behavior annunciation unit makes an annunciation member disposed so as to correspond to a behavior area where the subsequent behavior is to be executed operate in a first annunciation configuration based on the subsequent behavior determined by the determination unit. The correct behavior annunciation unit makes the annunciation member operate in a second annunciation configuration when the behavior recognition unit recognizes that the shopper executes the subsequent behavior. The warning unit which makes the annunciation member disposed so as to correspond to the behavior area where a different behavior from the subsequent behavior is executed operate in a warning configuration when the behavior recognition unit recognizes that the shopper executes the different behavior.

(18) Some embodiments of a control device will hereinafter be described using the drawings.

First Embodiment

(19) In the present embodiment, an explanation will be presented citing a settlement system of a self-service type as an example.

(20) FIG. 1 is a schematic configuration diagram showing the settlement system of a self-service type including a self-service POS terminal to which a behavior guide system according to a first embodiment including a control device according to the first embodiment is applied. The settlement system includes a plurality of self-service POS terminals **1** and a store server **2**. The settlement system connects the self-service POS terminals **1** and the store server **2** to each other with a communication network **3** such as a local area network (LAN). The store server **2** provides a variety of services related to the present settlement system to each of the self-service POS terminals **1** connected with the communication network **3**.

(21) The self-service POS terminals **1** as an example of the behavior guide system according to the first embodiment are installed in a payment place of a store, and a shopper who completes shopping is a user thereof. A customer who puts a purchased article into a shopping basket in a sales floor where articles are displayed, and then moves to the payment place operates the self-service POS terminal **1** to execute payment of the purchased article, namely a settlement related to a transaction from the registration of the article to the payment, by him- or herself. The self-service POS terminal **1** is an example of a settlement device which deals with a full-self-service settlement. The settlement device can be reworded as a settlement terminal, a payment device, a payment terminal, an article registration device, an article registration terminal, and so on.

(22) FIG. 2 is a perspective view showing an appearance configuration of the self-service POS terminal **1**, and FIG. 3 is a block diagram showing a circuit configuration of an essential part of the self-service POS terminal **1**.

(23) As shown in FIG. 2, the self-service POS terminal **1** is provided with a main body **110**

installed on a floor surface, and a weighing unit **112** disposed at the side of the main body **110**. The main body **110** is attached with a display pole **114**, a touch panel **116**, and two camera poles **118**, **120** in an upper part thereof. In the main body **110**, a basket table **122** is disposed in a central portion on a side surface at an opposite side to the side at which the weighing unit **112** is disposed. The basket table **122** is for the customer from the sales floor to place the basket containing the purchased article. The customer stands in front of the main body **110** in FIG. 2 so as to be able to see a screen of the touch panel **116** to perform an operation. Therefore, in a customer view, the basket table **122** is located at the right side, and the weighing unit **112** is located at the left side across the main body **110**. In the following description, a side at which the customer stands is defined as the front of the main body **110**, a side at which the weighing unit **112** is disposed is defined as the left side of the main body **110**, and a side at which the basket table **122** is disposed is defined as the right side of the main body **110**.

(24) The weighing unit **112** has a structure in which a weighing plate **126** is disposed in an upper part of a housing **124**, and a bag holding tool **128** is attached above the weighing plate **126**. An upper surface of the weighing plate **126** is defined as a placement surface **130**. The bag holding tool **128** is provided with a pair of holding arms **132**, and a plastic store bag or a shopping bag brought by the customer, a so-called own-bag, is held with these holding arms **132**. The weighing unit **112** measures the weight of the article placed which is put inside the plastic store bag or the own-bag held by the holding arms **132** and is placed on the placement surface **130**.

(25) The display pole **114** is provided with a light emitting unit **134** which selectively emits, for example, blue light and red light in a tip portion thereof. The display pole **114** displays the status of the self-service POS terminal **1** such as standby, in-operation, waiting for attendant, and halt with error using the colors of the light emitted by the light emitting unit **134**.

(26) The touch panel **116** is configured by a display for displaying a variety of screens to the user who operates the self-service POS terminal **1**, and a touch sensor for detecting touch input to the screen by the user. In the self-service POS terminal **1**, the user is normally a shopper.

(27) The camera pole **118** is attached with a basket area camera **136**, and the camera pole **120** is attached with a bagging area camera **138**.

(28) A mounting position and the posture of the basket area camera **136** are set so that the basket table **122** is included in the field angle, namely an imaging area, as indicated by dashed-dotted lines in FIG. 2. More specifically, the imaging area of the basket area camera **136** is set so that the inside of the shopping basket placed on the basket table **122** can be imaged from above. The imaging area of the basket area camera **136** is hereinafter referred to as a basket area.

(29) A mounting position and the posture of the bagging area camera **138** are set so that the placement surface **130** of the weighing plate **126** of the weighing unit **112** is included in the field angle, namely an imaging area, as indicated by dashed-dotted lines in FIG. 2. More specifically, the imaging area of the bagging area camera **138** is set so that an opening part for inputting an article of the plastic store bag or the own-bag held by the holding arms **132** can be imaged from above. The imaging area of the bagging area camera **138** is hereinafter referred to as a bagging area.

(30) In the placement surface **130** of the weighing plate **126**, a bagging area LED frame **140** for clearly showing the bagging area to a shopper is embedded. The bagging area LED frame **140** is provided with a plurality of green LED, a plurality of yellow LED, and a plurality of red LED, and is thus capable of presenting the bagging area to the shopper as a green frame, a yellow frame, or a red frame.

(31) It should be noted that although not particularly illustrated, it is possible to provide substantially the same LED frame also to the basket table **122**.

(32) A reading window **144** of a scanner **142** (see FIG. 3), a card slot **148** of a card reader **146** (see FIG. 3), an issue port **152** for a receipt printed by a printer **150** (see FIG. 3), and a speaker **154** are formed on a front surface of the main body **110**. A scanner area LED frame **156** is disposed so as to surround the periphery of the reading window **144** of the scanner **142**. Similarly to the bagging area

LED frame **140**, the scanner area LED frame **156** is provided with a plurality of green LED, a plurality of yellow LED, and a plurality of red LED, and is thus capable of presenting the reading window **144**, namely a scanner area, to the shopper as a green frame, a yellow frame, or a red frame. Further, on the front surface of the main body **110**, a coin slot **160** of an automatic change dispenser **158** (see FIG. 3), a coin dispensing port **162**, a bill slot **164**, and a bill dispensing port **166** are also formed.

(33) The bagging area LED frame **140** and the scanner area LED frame **156** are an example of the annunciation members disposed so as to correspond respectively to a plurality of behavior areas where a plurality of behaviors included in a series of behaviors to be executed by a shopper is executed. The bagging area and the scanner area are an example of the plurality of behavior areas. The bagging area LED frame **140** and the scanner area LED frame **156** are an example of the visual annunciation members and the light emitting members.

(34) Further, a communication cable **168** extends from a right side surface of the main body **110** to the outside, and a reader-and-writer **170** for an electronic medium is coupled to a tip of the communication cable **168**. The reader-and-writer **170** is placed on a placement table **172** provided to an upper portion on a right side surface of the main body **110**.

(35) As shown in FIG. 3, the self-service POS terminal **1** is provided with a processor **174**, a main memory **176**, an auxiliary storage device **178**, a time piece **180**, a communication interface **182**, and a system transmission path **184** in addition to the weighing unit **112**, the touch panel **116**, the light emitting unit **134**, the basket area camera **136**, the bagging area camera **138**, the bagging area LED frame **140**, the scanner **142**, the card reader **146**, the printer **150**, the speaker **154**, the scanner area LED frame **156**, the automatic change dispenser **158**, and the reader-and-writer **170** described above. The system transmission path **184** includes an address bus, a data bus, control signal lines, and so on. The system transmission path **184** couples the processor **174** and other units to each other directly, or via a signal input-and-output circuit, and transmits data signals which are mutually received or transmitted therebetween. The processor **174**, the main memory **176**, and the auxiliary storage device **178** are coupled to each other with the system transmission path **184** to thereby configure a computer of the self-service POS terminal **1**. The computer is an example of a control device according to the first embodiment which controls the annunciation members disposed so as to correspond respectively to the plurality of behavior areas where the plurality of behaviors included in the series of behaviors to be executed by a shopper is executed.

(36) The processor **174** corresponds to a central portion of the computer described above. The processor **174** controls each unit so as to realize a variety of types of functions as the self-service POS terminal **1** with an operating system or an application program. The processor **174** is, for example, a central processing unit (CPU). The processor **174** can also be, for example, a micro processing unit (MPU), a system on a chip (SoC), a digital signal processor (DSP), a graphics processing unit (GPU), an application specific integrated circuit (ASIC), a programmable logic device (PLD), or a field-programmable gate array (FPGA). Alternatively, the processor **174** can be what is obtained by combining some of these with each other.

(37) The main memory **176** corresponds to a main storage portion of the computer described above. The main memory **176** includes a nonvolatile memory area and a volatile memory area. The main memory **176** stores the operating system or the application program in the nonvolatile memory area. The main memory **176** stores data which is necessary for the processor **174** to execute processing for controlling each unit in the nonvolatile memory area or the volatile memory area in some cases. For example, the main memory **176** stores a transaction file **186**. The transaction file **186** is a data file for storing the data related to one commercial transaction to be processed by the self-service POS terminal **1**. The transaction file **186** stores data such as a transaction number, purchased article data, the total number of items, a total amount, a discount amount, and a settlement amount. The transaction numbers are series of numbers issued every time the commercial transaction is processed by the self-service POS terminal **1**. The purchased article data

is the record data generated for each of the articles traded in the commercial transaction identified by the transaction number. Here, the purchased article data is formed of items such as an article code, an article name, a price, the number of pieces, an amount. The number of pieces is the number of the purchased articles identified by the article code. The amount is the total amount corresponding to the number of purchased pieces. The transaction file **186** can store a plurality of purchased article data. The total number of pieces is the number of pieces obtained by combining the numbers of pieces in the respective purchased article data with each other. The total amount is an amount obtained by combining the amounts in the respective purchased article data with each other. The discount amount is a reduction amount with respect to the total amount. The settlement amount is an amount obtained by subtracting the discount amount from the total amount. The main memory **176** uses the volatile memory area as a work area in which data is rewritten as appropriate by the processor **174**. The nonvolatile memory area is, for example, a read only memory (ROM). The volatile memory area is, for example, a random access memory (RAM).

(38) The auxiliary storage device **178** corresponds to an auxiliary storage portion of the computer described above. For example, an electrically erasable programmable read-only memory (EEPROM), a hard disc drive (HDD), or a solid state drive (SSD) may be the auxiliary storage device **178**. The auxiliary storage device **178** stores data which is used by the processor **174** when performing a variety of types of processing, data which is generated by the processing in the processor **174**, and so on. For example, the auxiliary storage device **178** includes a behavior order storage unit **188** and an article image storage unit **190**. It should be noted that the auxiliary storage device **178** stores the application program described above in some cases.

(39) The behavior order storage unit **188** stores an order of the series of behaviors for the shopper to be required to perform with the self-service POS terminal **1**. The series of behaviors include, for example, a bag setting behavior of setting a bag such as the plastic store bag or the own-bag to the weighing unit **112**, a scanning behavior of scanning the code symbol attached to the article with the scanner **142** in order to perform the registration of the purchased article, and a bagging behavior of inputting the article thus scanned in the shopping bag, and the behavior order storage unit **188** stores the order of the behaviors in which order. The behavior order storage unit **188** is an example of a storage unit which stores the order of the plurality of behaviors to be performed when the shopper executes the registration of the purchased article.

(40) The article image storage unit **190** stores an appearance image of each of the articles to be sold in the store.

(41) The time piece **180** measures the time. The processor **174** performs the processing taking the time measured by the time piece **180** as the current time.

(42) The communication interface **182** performs data communication in compliance with a communication protocol set in advance with an external device which is connected via a network. The external device is, for example, the store server **2**.

(43) The scanner **142** reads the code symbol from the article put over the reading window **144**. To each of the articles to be sold in a store, the code symbol obtained by coding an article ID or the like for identifying that article is attached. The code symbol is, for example, a barcode. The code symbol may be, for example, a two-dimensional data code. The scanner **142** may be of a type which reads the code symbol by scanning with a laser beam, or of a type which reads the code symbol from an image taken by an imaging device.

(44) The card reader **146** reads card data recorded on a card medium such as a credit card or a reward card. The card reader **146** draws in the card medium inserted into the card slot **148** inside the main body **110** to read the card data, and then discharges the card medium from the card slot **148**.

(45) The printer **150** prints a receipt data representing a content of a business transaction or the like on a receipt form. The receipt form on which the receipt data is printed is discharged from an issue port **152**, cut by a cutter not shown, and then issued as a receipt or a voucher.

(46) The speaker **154** emits a voice message and an annunciation sound.

(47) The automatic change dispenser **158** includes a coin unit **192** and a bill unit **194**. The coin unit **192** sorts out coins dropped in a coin slot **160** one by one to identify the denomination, and then houses the coins in a safe by denomination. The coin unit **192** takes out the coins of the corresponding denomination from the safe based on, for example, change data, and then pays them out to a coin dispensing port **162**. The bill unit **194** sorts out bills inserted in the bill slot **164** one by one to identify the denomination, and then houses the bills in the safe by denomination. The bill unit **194** takes out the bills of the corresponding denomination from the safe based on, for example, change data, and then pays them out to the bill dispensing port **166**.

(48) The reader-and-writer **170** performs reading and rewriting of electronic money recorded on an electronic money medium. The electronic money medium is, for example, a noncontact IC card. The electronic money medium may also be an electronic device such as a smartphone or a tablet terminal.

(49) The operation of the self-service POS terminal **1** having such a configuration will hereinafter be described.

(50) FIG. **4** is a flowchart representing an example of a procedure of an essential part of information processing to be executed by the processor **174** of the self-service POS terminal **1**. It should be noted that the processing operation of the processor **174** shown in FIG. **4** makes the transition to ACT(n+1) after ACTn (n is a natural number) unless specifically described. Further, the procedure shown in FIG. **4** is illustrative only. The procedure is not particularly limited as long as substantially the same result can be obtained.

(51) As ACT**11**, the processor **174** awaits a start of an operation by the shopper. For example, when the processor **174** starts the operation in response to power activation, the processor **174** displays “please touch” or the like on the display of the touch panel **116**, and then gets into a standby state. In ACT**11**, whether the touch panel **116** was touched is determined. When a touch on the touch panel **116** by the shopper does not occur, the processor **174** determines NO in ACT**11**, and then repeats ACT**11**. When the touch panel **116** is touched by the shopper, the processor **174** determines YES in ACT**11**, and then proceeds to ACT**12**.

(52) As ACT**12**, the processor **174** awaits bag type selection by the shopper. For example, the processor **174** displays a selection screen regarding whether to buy the plastic store bag or to use the own-bag on the display of the touch panel **116**, and then receives the selection operation by the shopper. When the selection operation is performed, the processor **174** proceeds to ACT**13**. The processor **174** which executes the processing in ACT**12** is an example of a behavior recognition unit for recognizing the behavior executed by the shopper, and at the same time, an example of a determination unit for determining a subsequent behavior as a behavior which the shopper should subsequently perform based on the behavior recognized by the behavior recognition unit.

(53) As ACT**13**, the processor **174** lights the bagging area LED frame **140** with the yellow color as an attention-seeking color.

(54) As ACT**14**, the processor **174** outputs bag setting guidance. For example, the processor **174** performs a guidance display of “please set a bag” on the display of the touch panel **116**, and at the same time, outputs similar sound guidance from the speaker **154**. The processor **174** which executes the processing in ACT**13** is an example of a subsequent behavior annunciation unit which makes the annunciation member disposed so as to correspond to the behavior area where the subsequent behavior should be executed operate in a first annunciation configuration based on the subsequent behavior determined by the determination unit. The bagging area LED frame **140** is an example of the annunciation member disposed so as to correspond to the behavior area in which the subsequent behavior should be executed. The lighting with the yellow color is an example of an operation in the first annunciation configuration.

(55) Then, the processor **174** executes a behavior detection with respect to the bag setting behavior which is the first behavior in the series of behaviors stored in the behavior order storage unit **188**,

and in which the shopper sets a bag such as the plastic store bag or the own-bag to the weighing unit **112**.

(56) Specifically, first, as **ACT15**, the processor **174** obtains a bagging area image taken by the bagging area camera **138**.

(57) Then, as **ACT16**, the processor **174** determines whether a bag, namely the plastic store bag or the own-bag, is set to the weighing unit **112** based on the bagging area image. Specifically, the processor **174** recognizes an area of a hand of the shopper from the bagging area image to determine whether the hand entered the bagging area. For example, the processor **174** detects the position of the hand using a technique of an object detection such as a single shot multibox detector (SSD) known to the public, or a technique of a posture estimation such as OpenPose. It should be noted that regarding the bagging area, as described above, it is assumed that the bagging area camera **138** is installed so that the placement surface **130** is included in the field angle, namely the imaging area, thereof, but by detecting the bagging area in real time using the technique of the object detection such as SSD, it is possible to make it unnecessary to exactly set the mounting position and the posture of the bagging area camera **138**. When no bag is set, the processor **174** determines NO in **ACT16**, and then proceeds to **ACT15**. When the bag is set, the processor **174** determines YES in **ACT16**, and then proceeds to **ACT17**. It should be noted that in determining whether the bag is set, it is possible to use a change in value of a weight sensor or the like provided to the weighing unit **112**. The processor **174** which executes the processing in **ACT16** is an example of the behavior recognition unit for recognizing the behavior executed by the shopper.

(58) As **ACT17**, the processor **174** outputs correct behavior guidance. For example, the processor **174** outputs a sound effect representing a correct answer such as “ding-dong” from the speaker **154**.

(59) As **ACT18**, the processor **174** starts lighting the bagging area LED frame **140** which was lit with the yellow color in **ACT13** with the green color for a certain period of time. The certain period of time is not exactly regulated, but is sufficiently a period of time necessary for the shopper to start taking out the article located in the shopping basket placed on the basket table **122** from when the shopper set the bag. The certain period of time is, for example, about 1 second. The processor **174** which executes the processing in **ACT18** is an example of the correct behavior annunciation unit which makes the annunciation member operating in the first annunciation configuration operate in a second annunciation configuration different from the first annunciation configuration when the behavior recognition unit recognized that the shopper executed the subsequent behavior. The lighting with the green color for the certain period of time is an example of an operation in the second annunciation configuration.

(60) Then, the processor **174** executes the behavior detection regarding the scanning behavior of the shopper scanning the code symbol attached to the article with the scanner **142** in order to perform the registration of the purchased article and the bagging behavior of the shopper inputting the article thus scanned in the shopping bag as the subsequent behavior in the series of behaviors stored in the behavior order storage unit **188**.

(61) First, as **ACT19**, the processor **174** obtains a basket area image taken by the basket area camera **136**.

(62) As **ACT20**, the processor **174** determines whether the content of the shopping basket placed on the basket table **122** is empty from the basket area image using a known image recognition processing technique. When the content of the shopping basket is not empty, namely when an article not yet scanned remains, the processor **174** determines NO in **ACT20**, and then proceeds to **ACT21**. The processor **174** which executes the processing in **ACT20** is an example of the behavior recognition unit for recognizing the behavior executed by the shopper, and at the same time, an example of the determination unit for determining the subsequent behavior as the behavior which the shopper should subsequently perform based on the behavior recognized by the behavior recognition unit.

(63) As **ACT 21**, the processor **174** executes registration behavior processing as a substance of the

behavior detection with respect to the scanning behavior and the bagging behavior.

(64) FIG. 5 is a flowchart showing an example of a detailed procedure of the registration behavior processing in ACT21. The processor 174 first executes the behavior detection with respect to the scanning behavior.

(65) Specifically, first, as ACT2101, the processor 174 lights the scanner area LED frame 156 disposed on the periphery of the reading window 144 of the scanner 142 with the yellow color as the attention-seeking color. The processor 174 which executes the processing in ACT2101 is an example of the subsequent behavior annunciation unit which makes the annunciation member disposed so as to correspond to the behavior area where the subsequent behavior should be executed operate in the first annunciation configuration based on the subsequent behavior determined by the determination unit. The scanner area LED frame 156 is an example of the annunciation member disposed so as to correspond to the behavior area in which the subsequent behavior should be executed. The lighting with the yellow color is an example of the operation in the first annunciation configuration.

(66) As ACT2102, the processor 174 outputs article scanning guidance. For example, the processor 174 performs a guidance display of “please scan articles” on the display of the touch panel 116, and at the same time, outputs similar sound guidance from the speaker 154.

(67) As ACT2103, the processor 174 obtains the output of the scanner 142.

(68) As ACT2103, the processor 174 determines whether the article is scanned. When the code symbol attached to the article is scanned by the scanner 142, it is possible for the processor 174 to obtain the article code represented by the code symbol from the scanner 142. Therefore, depending on whether the article code is obtained, it is possible for the processor 174 to determine whether the article was scanned. When the article was scanned, the processor 174 determines YES in ACT2104, and then proceeds to ACT2109. When the article was not scanned, the processor 174 determines NO in ACT2104, and then proceeds to ACT2105. The processor 174 which executes the processing in ACT2104 is an example of the behavior recognition unit for recognizing the behavior executed by the shopper, and at the same time, an example of the determination unit for determining the subsequent behavior as the behavior which the shopper should subsequently perform based on the behavior recognized by the behavior recognition unit.

(69) As ACT2105, the processor 174 obtains the bagging area image taken by the bagging area camera 138.

(70) As ACT2106, the processor 174 determines whether the article was bagged based on the bagging area image. Specifically, it is possible for the processor 174 to determine whether the article was bagged by recognizing the area of the hand of the shopper from the bagging area image to determine whether the hand entered the bagging area. When the article is not bagged, the processor 174 determines NO in ACT2106, and then proceeds to ACT2103. The processor 174 which executes the processing in ACT2106 is an example of the behavior recognition unit for recognizing the behavior executed by the shopper.

(71) Due to such a processing loop in ACT2103 through ACT2106 as described above, the processor 174 gets into a standby state for scanning of articles. It should be noted that as long as the certain period of time described above does not elapse, the green lighting of the bagging area LED frame 140 continues in some cases during the standby state for scanning of articles.

(72) When the article which was not scanned is bagged during the standby state for scanning of articles, the processor 174 determines YES in ACT2106, and proceeds to ACT2107. As ACT2107, the processor 174 blinks the bagging area LED frame 140 with the red color as a warning color. The processor 174 which executes the processing in ACT2107 is an example a warning unit which makes the annunciation member disposed so as to correspond to the behavior area where a different behavior from the subsequent behavior is executed operate in a warning configuration different from the first and second annunciation configurations when the behavior recognition unit recognized that the shopper executed the different behavior. The blinking with the red color is an

example of the operation in the warning configuration.

(73) As ACT2108, the processor 174 outputs wrong behavior guidance. For example, the processor 174 outputs a warning voice such as “please scan the article before bagging” from the speaker 154, and at the same time, displays a similar warning message on the display of the touch panel 116 for a predetermined period of time. Subsequently, the processor 174 makes the transition to the processing in ACT2103.

(74) When the article was scanned during the standby state for scanning of articles in the processing loop in ACT2103 through ACT2106, the processor 174 determines YES in ACT2104, and proceeds to ACT2109. As ACT2109, the processor 174 executes article registration processing with the article code of the article thus obtained. The article registration processing is known to the public, and therefore, the description thereof will be omitted. Thus, the article sales data regarding the article thus scanned are stored in the transaction file 186.

(75) As ACT2110, the processor 174 outputs the correct behavior guidance. For example, the processor 174 outputs the sound effect representing the correct answer such as “ding-dong” from the speaker 154.

(76) As ACT2111, the processor 174 starts lighting the scanner area LED frame 156 which was lit with the yellow color in ACT2101, with the green color for a certain period of time. This certain period of time is not exactly regulated, and is sufficiently a time necessary for the shopper to start bagging the article which was scanned. The certain period of time is, for example, about 1 second. The processor 174 which executes the processing in ACT2111 is an example of the correct behavior annunciation unit which makes the annunciation member operating in the first annunciation configuration operate in the second annunciation configuration different from the first annunciation configuration when the behavior recognition unit recognized that the shopper executed the subsequent behavior. The lighting with the green color for the certain period of time is an example of the operation in the second annunciation configuration.

(77) Then, the processor 174 executes the behavior detection with respect to the bagging behavior. Specifically, first, as ACT2112, the processor 174 lights the bagging area LED frame 140 with the yellow color as the attention-seeking color. The processor 174 which executes the processing in ACT2112 is an example of the subsequent behavior annunciation unit which makes the annunciation member disposed so as to correspond to the behavior area where the subsequent behavior should be executed operate in the first annunciation configuration based on the subsequent behavior determined by the determination unit. The bagging area LED frame 140 is an example of the annunciation member disposed so as to correspond to the behavior area in which the subsequent behavior should be executed. The lighting with the yellow color is an example of the operation in the first annunciation configuration.

(78) As ACT2113, the processor 174 outputs bagging guidance. For example, the processor 174 performs a guidance display of “please bag [article name]” on the display of the touch panel 116, and at the same time, outputs similar sound guidance from the speaker 154. Here, [article name] is the article name of the article corresponding to the article code registered in ACT2109.

(79) As ACT2114, the processor 174 obtains the bagging area image taken by the bagging area camera 138.

(80) As ACT2115, the processor 174 determines whether the article named [article name] was bagged based on the bagging area image. Specifically, the processor 174 recognizes an area of a hand of the shopper from the bagging area image to determine whether the hand entered the bagging area. The method of detecting the position of the hand is as described in ACT16. Further, the processor 174 applies a known technique such as template matching to the image of the periphery of the hand to thereby determine whether the image of the article gripped by the hand coincides with the image of the article named [article name]. It should be noted that the images of the articles are stored in the article image storage unit 190 of the auxiliary storage device 178. In this way, it is possible for the processor 174 to determine bagging of the article named [article

name]. When the article named [article name] was bagged, the processor **174** determines YES in ACT**2115**, and then proceeds to ACT**2119**. When the article named [article name] is not bagged, the processor **174** determines NO in ACT**2115**, and then proceeds to ACT**2116**. The processor **174** which executes the processing in ACT**2115** is an example of the behavior recognition unit for recognizing the behavior executed by the shopper, and at the same time, an example of the determination unit for determining the subsequent behavior as the behavior which the shopper should subsequently perform based on the behavior recognized by the behavior recognition unit.

(81) As ACT**2116**, the processor **174** determines whether other articles than the article named [article name] are bagged. When other articles than the article named [article name] are also not bagged, the processor **174** determines NO in ACT**2116**, and then proceeds to ACT**2114**. The processor **174** which executes the processing in ACT**2116** is an example of the behavior recognition unit for recognizing the behavior executed by the shopper.

(82) Due to such a processing loop in ACT**2114** through ACT**2116** as described above, the processor **174** gets into a standby state for bagging of the article named [article name]. It should be noted that as long as the certain period of time described above does not elapse, the green lighting of the scanner area LED frame **156** continues in some cases during the standby state for bagging of articles.

(83) When an article other than the article named [article name] is bagged during the standby state for bagging of articles, the processor **174** determines YES in ACT**2116**, and proceeds to ACT**2117**. As ACT**2117**, the processor **174** blinks the bagging area LED frame **140** with the red color as the warning color. The processor **174** which executes the processing in ACT**2117** is an example of the warning unit which makes the annunciation member disposed so as to correspond to the behavior area where a different behavior from the subsequent behavior is executed operate in the warning configuration different from the first and second annunciation configurations when the behavior recognition unit recognized that the shopper executed the different behavior. The blinking with the red color is an example of the operation in the warning configuration.

(84) As ACT**2118**, the processor **174** outputs the wrong behavior guidance. For example, the processor **174** outputs a warning voice such as “please bag the scanned article” from the speaker **154**, and at the same time, displays a similar warning message on the display of the touch panel **116** for a predetermined period of time. Subsequently, the processor **174** makes the transition to the processing in ACT**2114**.

(85) As ACT**2119**, the processor **174** outputs the correct behavior guidance. For example, the processor **174** outputs the sound effect representing the correct answer such as “ding-dong” from the speaker **154**.

(86) As ACT**2120**, the processor **174** starts lighting the bagging area LED frame **140** which was lit with the yellow color in ACT**2112** with the green color for a certain period of time. The certain period of time is not exactly regulated, but is sufficiently a period of time necessary for the shopper to start taking out the next article located in the shopping basket placed on the basket table **122** from when the shopper bags the present article. The certain period of time is, for example, about 1 second. Subsequently, the processor **174** returns to the higher-level routine. The processor **174** which executes the processing in ACT**2120** is an example of the correct behavior annunciation unit which makes the annunciation member operating in the first annunciation configuration operate in the second annunciation configuration different from the first annunciation configuration when the behavior recognition unit recognized that the shopper executed the subsequent behavior. The lighting with the green color for the certain period of time is an example of the operation in the second annunciation configuration.

(87) It should be noted that ACT**18**, ACT**2111**, and ACT**2120** may be the same in certain period of time as each other, or may be different in certain period of time from each other.

(88) Referring back to the description of FIG. **4**, when the processing of the scanning behavior and the bagging behavior with respect to one article in the shopping basket as the registration behavior

processing in ACT21 is completed, the processor 174 proceeds to ACT19. In this way, during the processing loop in ACT19 through ACT21, the registration and the bagging with respect to each of the articles in the shopping basket by the shopper are executed as a result. When the registration and the bagging of all the articles in the shopping basket are completed, and thus, the content of the shopping basket becomes empty, the processor 174 determines YES in ACT20, and then proceeds to ACT22.

(89) As ACT22, the processor 174 executes payment processing based on the article sales data stored in the transaction file 186. The payment processing is known to the public, and therefore, the description thereof will be omitted. Thus, the shopper executes the payment of the purchased article.

(90) As described above, the computer including the processor 174 of the self-service POS terminal 1 is the control device according to the first embodiment which controls the annunciation members disposed so as to correspond respectively to the plurality of behavior areas where the plurality of behaviors included in the series of behaviors to be executed by the shopper is executed, and recognizes the behavior executed by the shopper to make the annunciation member disposed so as to correspond to the behavior area where the subsequent behavior is performed operate in accordance with the subsequent behavior which should subsequently be performed to thereby provide the shopper with the guidance on the behavior which should subsequently be performed, and where to perform the behavior. Then, when the subsequent behavior was correctly executed by the shopper, this fact is announced with that annunciation member, or when a wrong behavior was executed, warning is performed with the annunciation member disposed so as to correspond to the behavior area where the wrong behavior was executed. Therefore, the processor 174 configures the behavior recognition unit for recognizing the behavior executed by the shopper, the determination unit for determining the subsequent behavior as the behavior which the shopper should subsequently perform based on the behavior recognized by the behavior recognition unit, the subsequent behavior annunciation unit which makes the annunciation member disposed so as to correspond to the behavior area where the subsequent behavior should be executed operate in the first annunciation configuration based on the subsequent behavior determined by the determination unit, the correct behavior annunciation unit which makes the annunciation member operating in the first annunciation configuration operate in a second annunciation configuration different from the first annunciation configuration when the behavior recognition unit recognized that the shopper executed the subsequent behavior, and the warning unit which makes the annunciation member disposed so as to correspond to the behavior area where a different behavior from the subsequent behavior is executed operate in the warning configuration different from first and second annunciation the configurations when the behavior recognition unit recognized that the shopper executed the different behavior.

(91) As described above, according to the self-service POS terminal 1 including the control device related to the first embodiment, when the wrong behavior was executed as the subsequent behavior by the shopper, a warning against this fact is given to the shopper, and therefore, it becomes possible to prevent an involuntary erroneous behavior and a willful illicit behavior of the shopper.

(92) Further, the series of behaviors to be executed by the shopper include a plurality of behaviors which the shopper performs when executing the registration of the purchased article, the computer of the self-service POS terminal 1 includes the behavior order storage unit 188 which stores the order of the plurality of behaviors to be performed when executing the registration of the purchased article in the auxiliary storage device 178, and the processor 174 which configures the determination unit determines the subsequent behavior based on the order of the behaviors stored in the behavior order storage unit 188.

(93) Therefore, according to the self-service POS terminal 1 in the first embodiment, it is possible to easily determine the subsequent behavior to announce the subsequent behavior to the shopper.

(94) It should be noted that the plurality of behaviors includes at least the scanning behavior of

scanning the code symbol attached to the purchased article with the scanner **142**, and the processor **174** which configures the behavior recognition unit recognizes the fact that the shopper executed the scanning behavior based on the output of the scanner **142**.

(95) Therefore, according to the self-service POS terminal **1** in the first embodiment, it becomes possible to surely announce the registration behavior of the purchased article as an essential behavior to the shopper.

(96) Further, the annunciation members include visual annunciation members, and the first annunciation configuration, the second annunciation configuration, and the warning configuration by the processor **174** are different in visual annunciation configuration of the visual annunciation members from each other.

(97) Therefore, it becomes possible to make the shopper surely recognize the behavior area where the subsequent behavior should be performed, and whether the correct behavior was performed in that behavior area.

(98) It should be noted that the visual annunciation members each include the light emitting member as the LED frame, and the first annunciation configuration, the second annunciation configuration, and the warning configuration are different in at least one of color of the emitted light and lighting pattern from each other.

(99) Therefore, it is possible to make the annunciation configurations visually and clearly different from each other.

Second Embodiment

(100) Then, a second embodiment will be described. In the present embodiment, the description will be presented citing, as an example, a settlement system including a pre-registration system in which the customer him- or herself performs the article registration of a purchased article in advance of checkout in a payment machine using an information terminal which is attached to a shopping cart and is capable of reading an article code, or an information terminal which is provided with a scanning function for the customer him- or herself to perform reading of the article during shopping in a store. It should be noted that the configurations and operations substantially the same as those in the first embodiment will be attached with substantially the same reference symbols as in the first embodiment to thereby omit the description thereof.

(101) FIG. **6** is a schematic configuration diagram showing a self-service type settlement system including an information terminal **4** to which a behavior guide system according to the second embodiment including a control device according to the second embodiment is applied. The settlement system according to the second embodiment includes the information terminal **4**, an access point **5**, and a plurality of payment machines **6** in addition to the communication network **3**, the plurality of self-service POS terminals **1**, and the store server **2**. It should be noted that just one access point **5** is shown in FIG. **6**, but the number of the access points **5** may be two or more.

(102) The information terminal **4** is a terminal for the shopper to perform the article registration during shopping in the store. The access point **5** is a wireless base station for connecting the information terminal **4** to the communication network **3**. The information terminal **4** does not have the transaction file, and transmits the article code to the store server **2** via the access point **5** together with a terminal ID of that information terminal **4** to thereby store the transaction data in the transaction file for each of the information terminals **4**, namely each of the shoppers, provided to the store server **2**. It should be noted that it is also possible to connect another management device from the store server **2** to the communication network **3**, and manage the transaction file for each of the information terminals **4** in that management device. The payment machines **6** are each a checkout apparatus for the shopper to perform payment based on the transaction data for each of the information terminals **4** stored in the store server **2**. It should be noted that it is possible to provide the self-service POS terminal **1** with a function of the payment machine **6**.

(103) Further, it is possible for the settlement system to be provided with a plurality of sales floor monitoring cameras **7** connected to the communication network **3**. The sales floor monitoring

cameras **7** are disposed on, for example, the ceiling of each of the sales floors in the store, and shoot a condition in which the shopper scans an article arranged on a display shelf or the like, or input an article into a shopping basket mounted on a shopping cart from above the sales floor. A sales floor image taken by each of the sales floor monitoring cameras **7** is transmitted to the store server **2**, and is used for monitoring the sales floor. Further, it is also possible to arrange that the sales floor image can be transmitted to the information terminal **4** which is included in the field angle, namely the imaging area, of each of the sales floor monitoring cameras **7**. On this occasion, by installing a transmitter and a receiver of, for example, a radio field intensity with each of the access points **5** or a beacon in the sales floor and the information terminals **4** or the shopping cart, the store server **2** determines the location of each of the information terminals **4** to transmit the sales floor image of corresponding one of the sales floor monitoring cameras **7**.

(104) FIG. **7** is a perspective view showing an example of an appearance configuration of the information terminal **4** attached to the shopping cart **C**. Further, FIG. **8** is schematic diagram illustrating the imaging ranges of a surrounding area camera **412** and a basket area camera **414** which are provided to the information terminal **4**, and are configured separately from a main body **410** of the information terminal **4**.

(105) The information terminal **4** includes the main body **410** which is provided with a touch panel **416** and a speaker **418**, and which is attached with a card reader **420**, and a scanner **422** separated from the main body **410**.

(106) The shopping cart **C** is provided with a caster unit **C1** for moving, a handle frame unit **C2**, and a basket holder **C3**. The caster unit **C1** includes wheels (four wheels) for smoothly moving on a floor surface. The handle frame unit **C2** includes a pair of longitudinal frames **C21**, **C21** erected at a rear wheel side of the caster unit **C1**, and a handle bar **C22** which couples upper ends of these longitudinal frames **C21**, **C21**. The basket holder **C3** is located frontward from a middle region of the handle frame unit **C2**. The shopping cart **C** loads a shopping basket **BA** for housing an article on each of the basket holder **C3** and the caster unit **C1**.

(107) In the shopping cart **C**, a pole **C4** is attached to one of the longitudinal frames **C21**. A tip of the pole **C4** is located above the handle bar **C22**. The main body **410** is attached to a tip portion of the pole **C4** so that a screen of the touch panel **416** is located toward the rear side. The card reader **420** is attached to the main body **410** so that a card slit is located toward the back side. In FIG. **7**, a magnetic card reader is adopted as the card reader **420**. The scanner **422** is located at a middle portion of the handle bar **C22**. The scanner **422** is attached to the handle bar **C22** so that a reading window is located toward the front side.

(108) The surrounding area camera **412** is attached to, for example, an upper surface of the scanner **422**. The surrounding area camera **412** is, for example, an omnidirectional camera, and is attached so that all circumferential directions of the shopping cart **C** are included in the field angle, namely the imaging area, as indicated by dashed-two dotted lines in FIG. **8**. Thus, regardless of the positional relationship between the shopping cart **C** and the article which is picked up by the shopper from the article shelf, the condition thereof is imaged by the surrounding area camera **412**. The imaging area of the surrounding area camera **412** is hereinafter referred to as a surrounding area.

(109) As shown in FIG. **8**, the basket area camera **414** is attached to a rear surface of the main body **410** which is located at the front side of the shopping cart **C**. A mounting position and the posture of the basket area camera **414** are set so that the basket holder **C3** is included in the field angle, namely the imaging area, as indicated by dashed-dotted lines in FIG. **8**. More specifically, the imaging area of the basket area camera **414** is set so that the inside of the shopping basket **BA** housed in the basket holder **C3** can be imaged from obliquely above. The imaging area of the basket area camera **414** is hereinafter referred to as a basket area.

(110) A basket area LED frame **424** is attached to an upper surface of the basket holder **C3** so as to surround the shopping basket **BA** housed in the basket holder **C3**. Similarly to the bagging area

LED frame **140** in the first embodiment, the basket area LED frame **424** is provided with a plurality of green LED, a plurality of yellow LED, and a plurality of red LED, and is thus capable of presenting a bagging area to the shopper as a green frame, a yellow frame, or a red frame.

(111) Further, a scanner area LED frame **428** is disposed so as to surround the periphery of a reading window **426** of the scanner **422**. Similarly to the scanner area LED frame **156** in the first embodiment, the scanner area LED frame **428** is also provided with a plurality of green LED, a plurality of yellow LED, and a plurality of red LED, and is thus capable of presenting the reading window **426**, namely a scanner area, to the shopper as a green frame, a yellow frame, or a red frame.

(112) The basket area LED frame **424** and the scanner area LED frame **428** are an example of annunciation members disposed so as to correspond respectively to a plurality of behavior areas where a plurality of behaviors included in a series of behaviors to be executed by a shopper is executed. The basket holder **C3** and the scanner area are an example of the plurality of behavior areas. The basket area LED frame **424** and the scanner area LED frame **428** are an example of the visual annunciation members and the light emitting members.

(113) FIG. **9** is a block diagram showing an example of a circuit configuration of an essential part of the information terminal **4**. As shown in FIG. **9**, the information terminal **4** is provided with a processor **430**, a main memory **432**, an auxiliary storage device **434**, a wireless unit **436**, and a system transmission path **438** in addition to the surrounding area camera **412**, the basket area camera **414**, the touch panel **416**, the speaker **418**, the card reader **420**, the scanner **422**, the basket area LED frame **424**, and the scanner area LED frame **428** described above. The system transmission path **438** includes an address bus, a data bus, control signal lines, and so on. The system transmission path **438** couples the processor **430** and other units to each other directly, or via a signal input-and-output circuit, and transmits data signals which are mutually received or transmitted therebetween. The processor **430**, the main memory **432**, and the auxiliary storage device **434** are coupled to each other with the system transmission path **438** to thereby configure a computer of the information terminal **4**. The computer is an example of a control device according to the second embodiment which controls the annunciation members disposed so as to correspond respectively to the plurality of behavior areas where the plurality of behaviors included in the series of behaviors to be executed by a shopper is executed.

(114) The processor **430** corresponds to a central portion of the computer described above. The processor **430** controls each unit so as to realize a variety of types of functions as the information terminal **4** with an operating system or an application program. The processor **430** may be, for example, a CPU, an MPU, an SoC, a DSP, a GPU, and an ASIC, a PLD, or an FPGA similarly to the processor **174** of the self-service POS terminal **1**. Alternatively, the processor **430** can be what is obtained by combining some of these with each other.

(115) The main memory **432** corresponds to a main storage portion of the computer described above. The main memory **432** includes a nonvolatile memory area for storing the operating system or the application program, and a volatile memory area to be used as a work area where data is appropriately rewritten by the processor **430** similarly to the main memory **176** of the self-service POS terminal **1**. The main memory **432** stores data which is necessary for the processor **430** to execute processing for controlling each unit in the nonvolatile memory area or the volatile memory area in some cases. The nonvolatile memory area is, for example, a ROM, and the volatile memory area is, for example, a RAM.

(116) The auxiliary storage device **434** corresponds to an auxiliary storage portion of the computer described above. The auxiliary storage device **434** may be configured with, for example, an EEPROM, an HDD, or an SSD similarly to the auxiliary storage device **178** of the self-service POS terminal **1**, and stores data to be used by the processor **430** when performing a variety of types of processing, data generated by the processing of the processor **430**, and so on. It should be noted that the auxiliary storage device **434** stores the application program described above in some cases.

(117) Also in the present second embodiment, the auxiliary storage device **434** may include a behavior order storage unit **440** and an article image storage unit **442**. The behavior order storage unit **440** stores an order of the series of behaviors for the shopper to be required to perform with the information terminal **4**. The article image storage unit **442** stores an appearance image of each of the articles to be sold in the store.

(118) In the present embodiment, the series of behaviors stored by the behavior order storage unit **440** include, for example, a gripping behavior of the shopper picking up an article to be purchased, a scanning behavior of scanning the code symbol attached to an article with the scanner **422** in order to register the purchased article, and an input-to-basket behavior of inputting the scanned article into the shopping basket BA housed in the basket holder C3 of the shopping cart C, and the behavior order storage unit **440** stores the order of the behaviors in this order. The behavior order storage unit **440** is an example of a storage unit which stores the order of the plurality of behaviors to be performed when the shopper executes the registration of the purchased article

(119) The wireless unit **436** performs data communication in compliance with a communication protocol set in advance with an external device which is connected via a network by performing wireless communication with the access point **5**. The external device is, for example, the store server **2**.

(120) The scanner **422** reads the code symbol from the article put over the reading window **426**. The scanner **422** may be of a type which reads the code symbol by scanning with a laser beam, or of a type which reads the code symbol from an image taken by an imaging device.

(121) The card reader **420** reads card data recorded on a card medium such as a credit card or a reward card.

(122) The operation of the information terminal **4** having such a configuration will hereinafter be described. FIG. **10** is a flowchart representing an example of a procedure of an essential part of information processing to be executed by the processor **430** of the information terminal **4**.

(123) As ACT**41**, the processor **430** awaits a start of an operation by the shopper. For example, when the processor **430** starts the operation in response to power activation, the processor **174** displays “please touch” or the like on the display of the touch panel **416**, and then gets into a standby state. In ACT**41**, whether the touch panel **416** was touched is determined. When a touch on the touch panel **416** by the shopper does not occur, the processor **430** determines NO in ACT**41**, and then repeats ACT**41**. When the touch panel **416** is touched by the shopper, the processor **430** determines YES in ACT**41**, and then proceeds to ACT**42**.

(124) As ACT**42**, the processor **430** executes login processing. The login processing is processing of transmitting the member ID of the shopper read from the recording medium owned by the shopper with the card reader **420** to the store server **2** together with the terminal ID of that information terminal **4** with the wireless unit **436** via the access point **5** to get approval. The login processing is processing known to the public, and therefore, the detailed description thereof will be omitted. It should be noted that the store server **2** may generate a transaction file with respect to the member thus approved, update transaction data in the corresponding transaction file in accordance with the transmission of the article code attached with the terminal ID of the information terminal **4** from that information terminal **4**, and transmit the transaction data to the payment machine **6** to make the payment machine **6** perform checkout when the shopper checks out on the payment machine **6**.

(125) Further, the processor **430** executes a behavior detection with respect to the gripping behavior of the shopper picking up an article to be purchased by the shopper, which is the first behavior in the series of behaviors stored in the behavior order storage unit **440**.

(126) Specifically, as ACT**43**, the processor **430** obtains a surrounding image with respect to the surrounding area taken by the surrounding area camera **412**.

(127) As ACT**44**, the processor **430** determines whether the shopper grips an article displayed on the article shelf based on the surrounding area image. Specifically, the processor **430** recognizes an

area of a hand of the shopper from the surrounding area image to determine whether an object is gripped. The method of detecting the position of the hand is as described in the first embodiment. When no article is gripped, the processor **430** determines NO in ACT**44**, and then proceeds to ACT**45**. When an article is gripped, the processor **430** determines YES in ACT**44**, and then proceeds to ACT**46**. The processor **430** which executes the processing in ACT**44** is an example of a behavior recognition unit for recognizing the behavior executed by the shopper, and at the same time, an example of a determination unit for determining the subsequent behavior as the behavior which the shopper should subsequently perform based on the behavior recognized by the behavior recognition unit.

(128) As ACT**45**, the processor **430** determines whether a declaration for payment by the shopper was made. For example, a “payment” button to be used when the shopper declares the payment of the purchased article is displayed on the display of the touch panel **416**, and the shopper touches the “payment” button when making the transition to the payment. Therefore, it is possible for the processor **430** to determine the presence or absence of the payment declaration based on whether the “payment” button is touched. When the declaration of payment was made, the processor **430** determines YES in ACT**45**, and then proceeds to ACT**47**. When the declaration of payment was not made, the processor **430** determines NO in ACT**45**, and then proceeds to ACT**43**.

(129) Due to such a processing loop in ACT**43** through ACT**45** as described above, the processor **430** awaits gripping of an article or the payment declaration by the shopper. Then, when an article is gripped by the shopper, the processor **430** proceeds to ACT**46**.

(130) As ACT**46**, the processor **430** executes registration behavior processing as a substance of the behavior detection with respect to the scanning behavior of scanning the code symbol attached to the article with the scanner **422** in order to perform the registration of the purchased article, and the input-to-basket behavior of inputting the article thus scanned into the shopping basket BA housed in the basket holder C3 of the shopping cart C.

(131) FIG. **11** is a flowchart showing an example of a detailed procedure of the registration behavior processing in ACT**46**. The processor **430** first executes the behavior detection with respect to the scanning behavior.

(132) Specifically, first, as ACT**4601**, the processor **430** lights the scanner area LED frame **428** disposed on the periphery of the reading window **426** of the scanner **422** with the yellow color as the attention-seeking color. The processor **430** which executes the processing in ACT**4601** is an example of the subsequent behavior annunciation unit which makes the annunciation member disposed so as to correspond to the behavior area where the subsequent behavior should be executed operate in the first annunciation configuration based on the subsequent behavior determined by the determination unit. The scanner area LED frame **428** is an example of the annunciation member disposed as to correspond to the behavior area in which the subsequent behavior should be executed. The lighting with the yellow color is an example of the operation in the first annunciation configuration.

(133) As ACT**4602**, the processor **430** outputs article scanning guidance. For example, the processor **430** performs a guidance display of “please scan articles” on the display of the touch panel **416**, and at the same time, outputs similar sound guidance from the speaker **418**.

(134) As ACT**4603**, the processor **430** obtains the output of the scanner **422**.

(135) As ACT**4603**, the processor **430** determines whether the article was scanned, namely whether the article code represented by the code symbol was obtained from the scanner **422**. When the article was scanned, the processor **430** determines YES in ACT**4604**, and then proceeds to ACT**4609**. When the article was not scanned, the processor **430** determines NO in ACT**4604**, and then proceeds to ACT**4605**. The processor **430** which executes the processing in ACT**4604** is an example of the behavior recognition unit for recognizing the behavior executed by the shopper, and at the same time, an example of the determination unit for determining the subsequent behavior as the behavior which the shopper should subsequently perform based on the behavior recognized by

the behavior recognition unit.

(136) As ACT**4605**, the processor **430** obtains the basket area image taken by the basket area camera **414**.

(137) As ACT**4606**, the processor **430** determines whether the article was input into the shopping basket BA based on the basket area image. Specifically, it is possible for the processor **430** to determine whether the article was input into the shopping basket BA by recognizing the area of the hand of the shopper from the basket area image to determine whether the hand entered the basket area. When the article was not input into the shopping basket BA, the processor **430** determines NO in ACT**4606**, and then proceeds to ACT**4603**. The processor **430** which executes the processing in ACT**4606** is an example of the behavior recognition unit for recognizing the behavior executed by the shopper.

(138) Due to such a processing loop in ACT**4603** through ACT**4606** as described above, the processor **430** gets into a standby state for scanning of articles.

(139) When the article which was not scanned is input into the shopping basket BA during the standby state for scanning of articles, the processor **430** determines YES in ACT**4606**, and proceeds to ACT**4607**. As ACT**4607**, the processor **430** blinks the basket area LED frame **424** with the red color as the warning color. The processor **430** which executes the processing in ACT**4607** is an example of the warning unit which makes the annunciation member disposed so as to correspond to the behavior area where a different behavior from the subsequent behavior is executed operate in the warning configuration different from the first and second annunciation configurations when the behavior recognition unit recognized that the shopper executed the different behavior. The blinking with the red color is an example of the operation in the warning configuration.

(140) As ACT**4608**, the processor **430** outputs the wrong behavior guidance. For example, the processor **430** outputs a warning voice such as “please scan the article before inputting the article into the basket” from the speaker **418**, and at the same time, displays a similar warning message on the display of the touch panel **416** for a predetermined period of time. Subsequently, the processor **430** makes the transition to the processing in ACT**4603**.

(141) When the article was scanned during the standby state for scanning of articles in the processing loop in ACT**4603** through ACT**4606**, the processor **430** determines YES in ACT**4604**, and proceeds to ACT**4609**. As ACT**4609**, the processor **430** executes the article registration processing with the article code of the article thus obtained. In the article registration processing, as described above, the processor **430** transmits the article code thus obtained to the store server **2** with the wireless unit **436** via the access point **5**.

(142) As ACT**4610**, the processor **430** outputs the correct behavior guidance. For example, the processor **430** outputs the sound effect representing the correct answer such as “ding-dong” from the speaker **418**.

(143) As ACT**4611**, the processor **430** starts lighting the scanner area LED frame **428** which was lit with the yellow color in ACT**4601**, with the green color for a certain period of time. This certain period of time is not exactly regulated, and is sufficiently a time necessary for the shopper to start inputting the article which was scanned into the shopping basket BA. The certain period of time is, for example, about 1 second. The processor **430** which executes the processing in ACT**4611** is an example of the correct behavior annunciation unit which makes the annunciation member operating in the first annunciation configuration operate in the second annunciation configuration different from the first annunciation configuration when the behavior recognition unit recognized that the shopper executed the subsequent behavior. The lighting with the green color for the certain period of time is an example of the operation in the second annunciation configuration.

(144) Then, the processor **430** executes the behavior detection with respect to the input-to-basket behavior. Specifically, first, as ACT**4612**, the processor **430** lights the basket area LED frame **424** with the yellow color as the attention-seeking color. The processor **430** which executes the processing in ACT**4612** is an example of the subsequent behavior annunciation unit which makes

the annunciation member disposed so as to correspond to the behavior area where the subsequent behavior should be executed operate in the first annunciation configuration based on the subsequent behavior determined by the determination unit. The basket area LED frame **424** is an example of the annunciation member disposed so as to correspond to the behavior area in which the subsequent behavior should be executed. The lighting with the yellow color is an example of the operation in the first annunciation configuration.

(145) As ACT**4613**, the processor **430** outputs the input-to-basket guidance. For example, the processor **430** performs a guidance display of “please input [article name] into the basket” on the display of the touch panel **416**, and at the same time, outputs similar sound guidance from the speaker **418**. Here, [article name] is the article name of the article corresponding to the article code registered in ACT**4609**.

(146) As ACT**4614**, the processor **430** obtains the basket area image taken by the basket area camera **414**.

(147) As ACT**4615**, the processor **430** determines whether the article named [article name] was input into the shopping basket BA based on the basket area image. Specifically, the processor **430** recognizes an area of the hand of the shopper from the basket area image to determine whether the hand entered the basket area. The method of detecting the position of the hand is as described in ACT**44**. Further, the processor **430** applies a known technique such as template matching to the image of the periphery of the hand to thereby determine whether the image of the article gripped by the hand coincides with the image of the article named [article name]. It should be noted that the images of the articles are stored in the article image storing unit **442** of the auxiliary storage device **434**. In this way, it is possible for the processor **430** to determine the input of the article named [article name] into the shopping basket BA. When the article named [article name] was input into the shopping basket BA, the processor **430** determines YES in ACT**4615**, and then proceeds to ACT**4619**. When the article named [article name] is not input into the shopping basket BA, the processor **430** determines NO in ACT**4615**, and then proceeds to ACT**4616**. The processor **430** which executes the processing in ACT**4615** is an example of the behavior recognition unit for recognizing the behavior executed by the shopper, and at the same time, an example of the determination unit for determining the subsequent behavior as the behavior which the shopper should subsequently perform based on the behavior recognized by the behavior recognition unit.

(148) As ACT**4616**, the processor **430** determines whether other articles than [article name] were input into the shopping basket BA. When other articles than the article named [article name] are also not input into the shopping basket BA, the processor **430** determines NO in ACT**4616**, and then proceeds to ACT**4614**. The processor **430** which executes the processing in ACT**4616** is an example of the behavior recognition unit for recognizing the behavior executed by the shopper.

(149) Due to such a processing loop in ACT**4614** through ACT**4616** as described above, the processor **430** gets into a standby state for the input of the article named [article name] into the shopping basket BA. It should be noted that as long as it is until the certain period of time described above elapses, the green lighting of the scanner area LED frame **428** continues in some cases during the standby state for inputting the article into the basket.

(150) When an article other than the article named [article name] is input into the shopping basket BA during the standby state for inputting the article named [article name] into the basket, the processor **430** determines YES in ACT**4616**, and proceeds to ACT**4617**. As ACT**4617**, the processor **430** blinks the basket area LED frame **424** with the red color as the warning color. The processor **430** which executes the processing in ACT**4617** is an example of the warning unit which makes the annunciation member disposed so as to correspond to the behavior area where a different behavior from the subsequent behavior is executed operate in the warning configuration different from the first and second annunciation configurations when the behavior recognition unit recognized that the shopper executed the different behavior. The blinking with the red color is an example of the operation in the warning configuration.

(151) As ACT**4618**, the processor **430** outputs the wrong behavior guidance. For example, the processor **430** outputs a warning voice such as “please input the scanned article into the basket” from the speaker **418**, and at the same time, displays a similar warning message on the display of the touch panel **416** for a predetermined period of time. Subsequently, the processor **430** makes the transition to the processing in ACT**4614**.

(152) As ACT**4619**, the processor **430** outputs the correct behavior guidance. For example, the processor **430** outputs the sound effect representing the correct answer such as “ding-dong” from the speaker **418**.

(153) As ACT**4620**, the processor **430** starts lighting the basket area LED frame **424** which was lit with the yellow color in ACT**4612** with the green color for a certain period of time. The certain period of time is not exactly regulated, but can be set to, for example, about 1 second similarly to other certain period of time. Subsequently, the processor **430** returns to the higher-level routine. The processor **430** which executes the processing in ACT**4620** is an example of the correct behavior annunciation unit which makes the annunciation member operating in the first annunciation configuration operate in the second annunciation configuration different from the first annunciation configuration when the behavior recognition unit recognized that the shopper executed the subsequent behavior. The lighting with the green color for the certain period of time is an example of the operation in the second annunciation configuration.

(154) It should be noted that ACT**4611** and ACT**4620** may be the same in certain period of time as each other, or may be different in certain period of time from each other.

(155) Referring back to the description of FIG. **10**, when the processing of the scanning behavior and the input-to-basket behavior with respect to one article in the shopping basket as the registration behavior processing in ACT**46** is completed, the processor **430** proceeds to ACT**15**. In this way, during the processing loop in ACT**43** through ACT**46**, the registration of the articles and the input of the articles into the shopping basket BA by the shopper are executed as a result. Then, when the payment is declared by the shopper, the processor **430** determines YES in ACT**45**, and then proceeds to ACT**47**.

(156) As ACT**47**, the processor **430** executes the payment processing. In this payment processing, for example, the processor **430** transmits a payment command including the terminal ID of that information terminal **4** to the store server **2** with the wireless unit **436** via the access point **5**. In the store server **2**, a unique payment code is issued in response to the reception of this payment command, and then, sends back a barcode image representing the payment code to that information terminal **4**. The shopper makes the display of the touch panel **416** display this barcode image to allow the scanner provided to the payment machine **6** to read the barcode image. Thus, it is possible for the payment machine **6** to obtain the payment code, and obtain the transaction data corresponding to the payment code from the store server **2**, and thus executes the checkout processing.

(157) As described above, it is possible to adopt the computer including the processor **430** of the information terminal **4** as the control device. Therefore, according also to the present second embodiment, substantially the same advantages as those of the first embodiment can be obtained.

(158) It should be noted that although the description is presented citing the information terminal **4** attached to the shopping cart C as an example, it is possible to adopt a smartphone owned by the shopper as the control device of the information terminal **4**. In this case, an application program for executing the function as the control device according to the present embodiment is installed in the smartphone. The shopper uses the shopping cart C provided with the basket area LED frame **424**. Scanning of an article is executed using a camera provided to the smartphone, and it is possible to use the sales floor monitoring cameras **7** as the cameras to be used for the behavior recognition of the shopper. The smartphone obtains the sales floor image from the sales floor monitoring camera **7** corresponding to the position where that smartphone is located via the access point **5**, and extracts an area image as a pixel portion of the area where the behavior detection should be performed as

the subsequent behavior from the sales floor image thus obtained. It is possible to extract the area image based on the position of the hand of the shopper in the sales floor layout or the position of the hand with respect to each unit of the shopping cart C detected in real time using a technique of an object detection such as SSD. Since the smartphone does not have the scanner area LED frame **428**, it is assumed that a scan message, an image showing the condition of performing scanning, or the like is displayed on the display. When performing the display, the color, the display pattern, or the color and the display pattern can be changed in accordance with whether the behavior is correct or wrong.

Third Embodiment

(159) Then, a third embodiment will be described. It should be noted that the configurations and operations substantially the same as those in the first embodiment will be attached with substantially the same reference symbols as in the first embodiment to thereby omit the description thereof.

(160) In the first and second embodiments described above, there are respectively described the behavior guide systems in which the annunciation members and the control device are provided to the self-service POS terminal **1** and the information terminal **4**, but it is possible to adopt a behavior guide system in which the annunciation members and the control device are provided to respective devices separated from each other.

(161) FIG. **12** is a schematic configuration diagram showing a self-service type settlement system as a behavior guide system according to the third embodiment including the store server **2** as the control device according to the third embodiment. The settlement system according to the present embodiment includes a plurality of checkout monitoring cameras **8** in addition to the plurality of self-service POS terminals **1** and the store server **2**.

(162) FIG. **13** is a perspective view showing an example of an imaging range of the checkout monitoring camera **8**. The checkout monitoring cameras **8** are disposed so as to correspond one-to-one to the self-service POS terminals **1**, and are installed on the ceiling or the like above the self-service POS terminals **1**, respectively. The mounting position and the posture of the checkout monitoring camera **8** are set so that the field angle, namely the imaging area, includes the basket area and the bagging area in the first embodiment as indicated by the dashed-dotted lines in FIG. **13**.

(163) The self-service POS terminal **1** in the present embodiment is not provided with the basket area camera **136** and the bagging area camera **138**, but the bagging area LED frame **140** and the scanner area LED frame **156** as the annunciation members are disposed similarly to the first embodiment.

(164) FIG. **14** is a sequence diagram illustrating an operation of such a settlement system.

(165) When the operation is started by the shopper and the bag selection is performed as in ACT**11** and ACT**12** described in the first embodiment, the self-service POS terminal **1** transmits (ACT**101**) a start notice to the store server **2**.

(166) The store server **2** determines the subsequent behavior as the behavior which the shopper should subsequently perform in response to the start notice, and then transmit (ACT**102**) a notification-and-lighting command to the self-service POS terminal **1** based on the subsequent behavior thus determined. It should be noted that the operations performed by the store server **2** in this sequence chart are executed by a processor provided to the store server **2** in reality, but there is simply described here that the operations are performed by the store server **2**. The notification-and-lighting command may be, for example, a command of lighting the bagging area LED frame **140** of the self-service POS terminal **1** with the yellow color as the attention-seeking color, and a command of making the display of the touch panel **116** perform the guidance display of “please set a bag,” and at the same time, outputting similar sound guidance from the speaker **154**. Therefore, the store server **2** is an example of the behavior recognition unit for recognizing the behavior executed by the shopper, and at the same time, an example of the determination unit for

determining the subsequent behavior as the behavior which the shopper should subsequently perform based on the behavior recognized by the behavior recognition unit. Further, the store server **2** is an example of the subsequent behavior annunciation unit which makes the annunciation member disposed so as to correspond to the behavior area where the subsequent behavior should be executed operate in the first annunciation configuration based on the subsequent behavior determined by the determination unit.

(167) In response to the notification-and-lighting command from the store server **2**, the processor **174** of the self-service POS terminal **1** lights the bagging area LED frame **140** with the yellow color as the attention-seeking color, performs the guidance display on the display of the touch panel **116**, and at the same time, outputs the sound guidance from the speaker **154** as in ACT**13** and ACT**14** in the first embodiment (ACT**103**). The bagging area LED frame **140** is an example of the annunciation member disposed so as to correspond to the behavior area in which the subsequent behavior should be executed. The lighting with the yellow color is an example of the operation in the first annunciation configuration.

(168) Further, the store server **2** obtains (ACT**104**) a monitoring image taken by the checkout monitoring camera **8** corresponding to the self-service POS terminal **1**.

(169) Then, the store server **2** determines (ACT**105**) the behavior of the shopper based on the monitoring image. Specifically, the store server **2** first extracts an area image as the pixel portion of an area where the behavior detection as the subsequent behavior should be performed from the monitoring image thus obtained. For example, when the subsequent behavior is a behavior related to the bag setting behavior of setting a bag such as the plastic store bag or the own-bag to the weighing unit **112**, the store server **2** extracts the pixel portion corresponding to such a bagging area as described in the first embodiment as the bagging area image from the monitoring image. This is an operation corresponding to ACT**15** in the first embodiment. Then, the store server **2** recognizes the behavior which the shopper currently executes based on the bagging area image thus extracted. The specific method of the recognition is as described in the first embodiment.

Therefore, the store server **2** is an example of the behavior recognition unit for recognizing the behavior executed by the shopper. Then, the store server **2** determines whether the behavior of setting a bag was performed. This is an operation corresponding to ACT**16** in the first embodiment.

(170) When the behavior of setting a bag was performed, the store server **2** transmits (ACT**106**) the notification-and-lighting command to the self-service POS terminal **1**. This notification-and-lighting command may be, for example, a command of lighting the bagging area LED frame **140** of the self-service POS terminal **1** with the green color for a certain period of time, and a command of outputting a sound effect representing a correct answer such as “ding-dong” from the speaker **154**. Therefore, the store server **2** is an example of the correct behavior annunciation unit which makes the annunciation member operating in the first annunciation configuration operate in the second annunciation configuration different from the first annunciation configuration when the behavior recognition unit recognized that the shopper executed the subsequent behavior. Further, as the notification-and-lighting command, it is possible to continue a command of lighting the scanner area LED frame **156** arranged on the periphery of the reading window **144** of the scanner **142** of the self-service POS terminal **1** with the yellow color as the attention-seeking color, and a command of making the display of the touch panel **116** perform the guidance display of “please scan the article,” and at the same time, outputting similar sound guidance from the speaker **154**. Therefore, the store server **2** is an example of the subsequent behavior annunciation unit which makes the annunciation member disposed so as to correspond to the behavior area where the subsequent behavior should be executed operate in the first annunciation configuration based on the subsequent behavior determined by the determination unit.

(171) In response to the notification-and-lighting command from the store server **2**, the processor **174** of the self-service POS terminal **1** starts lighting the bagging area LED frame **140** with the green color for a certain period of time, and at the same time, outputs the sound effect representing

a correct answer from the speaker **154** as in ACT**17** and ACT**18** in the first embodiment (ACT**107**). The lighting with the green color for the certain period of time is an example of the operation in the second annunciation configuration. Further, the processor **174** of the self-service POS terminal **1** lights the scanner area LED frame **156** with the yellow color as the attention-seeking color, performs the guidance display on the display of the touch panel **116**, and at the same time, outputs the sound guidance from the speaker **154** as in ACT**2101** and ACT**2102** in the first embodiment. The scanner area LED frame **156** is an example of the annunciation member disposed so as to correspond to the behavior area in which the subsequent behavior should be executed. The lighting with the yellow color is an example of the operation in the first annunciation configuration.

(172) Then, the store server **2** obtains (ACT**108**) the monitoring image taken by the checkout monitoring camera **8** corresponding to the self-service POS terminal **1**.

(173) Then, the store server **2** makes the completion determination, namely determines whether the content of the shopping basket is empty (ACT**109**). Specifically, the pixel portion corresponding to such a basket area as described in the first embodiment is extracted as the basket area image from the monitoring image thus obtained. This is an operation corresponding to ACT**19** in the first embodiment. Then, the store server **2** determines whether the content of the shopping basket is empty from the basket area image thus extracted using a known image recognition processing technique. This is an operation corresponding to ACT**20** in the first embodiment. Therefore, the store server **2** is an example of the behavior recognition unit for recognizing the behavior executed by the shopper, and at the same time, an example of the determination unit for determining the subsequent behavior as the behavior which the shopper should subsequently perform based on the behavior recognized by the behavior recognition unit.

(174) When the content of the shopping basket is not empty, the store server **2** obtains (ACT**1010**) the scanning result of the scanner **142** from the self-service POS terminal **1**. This is an operation corresponding to ACT**2103** in the first embodiment. Further, the store server **2** obtains the monitoring image taken by the checkout monitoring camera **8** corresponding to the self-service POS terminal **1**.

(175) Then, the store server **2** determines (ACT**1011**) the behavior of the shopper based on the scanning result and the monitoring image. Specifically, the store server **2** determines whether the article scan was performed based on the scanning result thus obtained. This is an operation corresponding to ACT**2104** in the first embodiment. Further, the store server **2** extracts the pixel portion corresponding to such a bagging area as described in the first embodiment as the bagging area image from the monitoring image thus obtained, and then determines whether the article was bagged based on the bagging area image. This is an operation corresponding to ACT**2104** and ACT**2105** in the first embodiment. Therefore, the store server **2** is an example of the behavior recognition unit for recognizing the behavior executed by the shopper, and at the same time, an example of the determination unit for determining the subsequent behavior as the behavior which the shopper should subsequently perform based on the behavior recognized by the behavior recognition unit.

(176) When the behavior of scanning the article was performed, or when the behavior of bagging the article was performed, the store server **2** transmits (ACT**1012**) the notification-and-lighting command to the self-service POS terminal **1**.

(177) When the article scan was performed, this notification-and-lighting command may be a command of lighting the bagging area LED frame **140** of the self-service POS terminal **1** with the green color for a certain period of time, and a command of outputting a sound effect representing a correct answer such as “ding-dong” from the speaker **154**. Therefore, the store server **2** is an example of the correct behavior annunciation unit which makes the annunciation member operating in the first annunciation configuration operate in the second annunciation configuration different from the first annunciation configuration when the behavior recognition unit recognized that the shopper executed the subsequent behavior. Further, as the notification-and-lighting command, it is

possible to continue a command of lighting the bagging area LED frame **140** of the self-service POS terminal **1** with the yellow color as the attention-seeking color, and a command of making the display of the touch panel **116** perform the guidance display of “please bag [article name],” and at the same time, outputting similar sound guidance from the speaker **154**. Therefore, the store server **2** is an example of the subsequent behavior annunciation unit which makes the annunciation member disposed so as to correspond to the behavior area where the subsequent behavior should be executed operate in the first annunciation configuration based on the subsequent behavior determined by the determination unit.

(178) Further, when an article which was not scanned was bagged, the notification-and-lighting command may be a command of blinking the bagging area LED frame **140** of the self-service POS terminal **1** with the red color as the warning color, and a command of outputting a warning voice such as “please scan the article before bagging” from the speaker **154**, and at the same time, displaying a similar warning message on the display of the touch panel **116** for a predetermined period of time. Therefore, the store server **2** is an example of the warning unit which makes the annunciation member disposed so as to correspond to the behavior area where a different behavior from the subsequent behavior is executed operate in the warning configuration different from the first and second annunciation configurations when the behavior recognition unit recognized that the shopper executed the different behavior. The blinking with the red color is an example of the operation in the warning configuration.

(179) In response to the notification-and-lighting command from the store server **2**, the processor **174** of the self-service POS terminal **1** performs (ACT**1013**) the sound output and the lighting based on the command.

(180) Specifically, the processor **174** starts lighting the scanner area LED frame **156** with the green color for a certain period of time, and then outputs the sound effect representing a correct answer from the speaker **154** as in ACT**2110** and ACT**2111** in the first embodiment. The lighting with the green color for the certain period of time is an example of the operation in the second annunciation configuration. Further, the processor **174** lights the bagging area LED frame **140** with the yellow color as the attention-seeking color, performs the guidance display on the display of the touch panel **116**, and at the same time, outputs the sound guidance from the speaker **154** as in ACT**2112** and ACT**2123** in the first embodiment. The bagging area LED frame **140** is an example of the annunciation member disposed so as to correspond to the behavior area in which the subsequent behavior should be executed. The lighting with the yellow color is an example of the operation in the first annunciation configuration.

(181) Alternatively, the processor **174** blinks the bagging area LED frame **140** with the red color as the warning color, outputs the warning voice from the speaker **154**, and at the same time, displays a similar warning message on the display of the touch panel **116** for a predetermined period of time as in ACT**2107** and ACT**2108** in the first embodiment. The blinking with the red color is an example of the operation in the warning configuration.

(182) Further, when the article scan was performed, the store server **2** obtains (ACT**1014**) the monitoring image taken by the checkout monitoring camera **8** corresponding to the self-service POS terminal **1**.

(183) Then, the store server **2** determines (ACT**1015**) the behavior of the shopper based on the monitoring image. Specifically, the store server **2** extracts the pixel portion corresponding to such a bagging area as described in the first embodiment as the bagging area image from the monitoring image thus obtained, and then determines whether the article named [article name] or an article other than the article named [article mane] was bagged based on the bagging area image. This is an operation corresponding to ACT**2114**, ACT**2115**, and ACT**2116** in the first embodiment. Therefore, the store server **2** is an example of the behavior recognition unit for recognizing the behavior executed by the shopper, and at the same time, an example of the determination unit for determining the subsequent behavior as the behavior which the shopper should subsequently

perform based on the behavior recognized by the behavior recognition unit.

(184) When the behavior of bagging any of the articles was performed, the store server **2** transmits (ACT**1016**) the notification-and-lighting command to the self-service POS terminal **1**.

Subsequently, the store server **2** proceeds to the operation in ACT**108**.

(185) When the article scan of [article name] was performed, this notification-and-lighting command may be a command of lighting the bagging area LED frame **140** of the self-service POS terminal **1** with the green color for a certain period of time, and a command of outputting a sound effect representing a correct answer such as “ding-dong” from the speaker **154**. Therefore, the store server **2** is an example of the correct behavior annunciation unit which makes the annunciation member operating in the first annunciation configuration operate in the second annunciation configuration different from the first annunciation configuration when the behavior recognition unit recognized that the shopper executed the subsequent behavior. Further, as the notification-and-lighting command, it is possible to continue a command of lighting the scanner area LED frame **156** of the self-service POS terminal **1** with the yellow color as the attention-seeking color, and a command of making the display of the touch panel **116** perform the guidance display of “please scan the article,” and at the same time, outputting similar sound guidance from the speaker **154**. Therefore, the store server **2** is an example of the subsequent behavior annunciation unit which makes the annunciation member disposed so as to correspond to the behavior area where the subsequent behavior should be executed operate in the first annunciation configuration based on the subsequent behavior determined by the determination unit.

(186) Further, when the article scan other than [article name] was performed, the notification-and-lighting command may be a command of blinking the bagging area LED frame **140** of the self-service POS terminal **1** with the red color as the warning color, and a command of outputting a warning voice such as “please bag the scanned article” from the speaker **154**, and at the same time, displaying a similar warning message on the display of the touch panel **116** for a predetermined period of time. Therefore, the store server **2** is an example of the warning unit which makes the annunciation member disposed so as to correspond to the behavior area where a different behavior from the subsequent behavior is executed operate in the warning configuration different from the first and second annunciation configurations when the behavior recognition unit recognized that the shopper executed the different behavior. The blinking with the red color is an example of the operation in the warning configuration.

(187) In response to the notification-and-lighting command from the store server **2**, the processor **174** of the self-service POS terminal **1** performs (ACT**1017**) the sound output and the lighting based on the command.

(188) Specifically, the processor **174** starts lighting the bagging area LED frame **140** with the green color for a certain period of time, and then outputs the sound effect representing a correct answer from the speaker **154** as in ACT**2119** and ACT**2120** in the first embodiment. The lighting with the green color for the certain period of time is an example of the operation in the second annunciation configuration. Further, the processor **174** lights the scanner area LED frame **156** with the yellow color as the attention-seeking color, performs the guidance display on the display of the touch panel **116**, and at the same time, outputs the sound guidance from the speaker **154** as in ACT**2101** and ACT**2102** in the first embodiment. The scanner area LED frame **156** is an example of the annunciation member disposed so as to correspond to the behavior area in which the subsequent behavior should be executed. The lighting with the yellow color is an example of the operation in the first annunciation configuration.

(189) Alternatively, the processor **174** blinks the bagging area LED frame **140** with the red color as the warning color, outputs the warning voice from the speaker **154**, and at the same time, displays a similar warning message on the display of the touch panel **116** for a predetermined period of time as in ACT**2117** and ACT**2118** in the first embodiment. The blinking with the red color is an example of the operation in the warning configuration.

(190) Further, in ACT**109**, when termination is determined, namely it is determined that the content of the shopping basket is empty, the store server **2** transmits (ACT**1018**) a payment command to the self-service POS terminal **1**.

(191) In response to the payment command from the store server **2**, the processor **174** of the self-service POS terminal **1** executes (ACT**1019**) the payment processing based on the article sales data stored in the transaction file **186**.

(192) As described above, it is possible to use the store server **2** as the control device. Therefore, according also to the present third embodiment, substantially the same advantages as those of the first embodiment can be obtained.

(193) It should be noted that the above description is presented using an example corresponding to the first embodiment, but it is obvious that the present embodiment can be made to correspond also to the second embodiment.

(194) Further, there is described the example in which the store server **2** acts as the control device, but it is obvious that it is possible for the management server disposed on the cloud outside the store to act as the control device.

Fourth Embodiment

(195) The behavior guide system corresponding to the series of behaviors to be executed by the shopper is described as the first through third embodiments, but the present disclosure can also be applied to behaviors other than shopping. Further, it is assumed that what behavior was performed is recognized based on the images taken by the camera, but it is possible to execute the behavior detection using a sensor other than the camera.

(196) For example, as a behavior regarded as important on the grounds of the recent spread of infection, there can be cited a hand-wash behavior.

(197) FIG. **15** is a plan view showing an example of an appearance configuration of a hand-wash system to which a behavior guide system according to the fourth embodiment including a control device according to the fourth embodiment is applied. The hand-wash unit includes a wash-hand basin **W**. The wash-hand basin **W** is provided with an automatic water faucet **W1** from which water is discharged, an automatic soap faucet **W2** from which soapy water is discharged, and an automatic blower faucet **W3** from which warm air blows off, and object detection sensors **W4** for detecting that a hand is located at a predetermined position are disposed in order to make the respective faucets operate.

(198) In the present embodiment, a water area LED frame **92** is disposed at the root of the automatic water faucet **W1** in the wash-hand basin **W**, a soap area LED frame **94** is similarly disposed at the root of the automatic soap faucet **W2**, and a blower area LED frame **96** is similarly disposed at the root of the automatic blower faucet **W3**.

(199) Further, a control device not shown defines a wash-with-water behavior, a wash-with-soap behavior, and a blower behavior as a series of hand-wash behavior, stores an order of the wash-with-water behavior and the wash-with-soap behavior including the number of times and durations thereof, and appropriately lights the LED frames **92**, **94**, and **96**.

(200) In the first through third embodiments, it is possible to use a detection result of the hand of the shopper with a sensor such as the object detection sensor **W4** when recognizing any of the behaviors.

Other Embodiments

(201) The embodiments of the control device are hereinabove described, but the embodiments are not limited thereto.

(202) For example, the light emitting members which function as the annunciation members are not required to be formed to have the frame shape surrounding the behavior area. For example, it is possible to dispose the green LED, the yellow LED, and the red LED one by one at any one of the places in the behavior area.

(203) Further, the annunciation member is not limited to the light emitting member, and can also be

a projection device for projecting a spot light, or can also be a message or an image to be displayed on the display of the touch panel **116, 416**.

(204) The scanner **142, 422** is not limited to what scans the code symbol attached to the article to output the article code, and can be a camera for imaging an appearance of the article, and it is possible to arrange that the processor **174, 430** identifies the article from the article image using a pattern matching technique. In this case, the scanning behavior described in the embodiments is replaced with an imaging behavior.

(205) Further, it is assumed in the first embodiment that the bagging area camera **138** is installed so that the placement surface **130** is included in the field angle, namely the imaging area, thereof, but it is possible to install that camera so that a wider range shows up. In this case, as described in the second embodiment, by extracting the pixel portion corresponding to the bagging area from the taken image of the bagging area camera **138**, it is possible to obtain the bagging area image.

(206) Further, the flow of the information processing described above to be performed by the processors **174, 430** is illustrative only, and is not limited to this order. For example, it is possible to exchange the order of ACT**13** and ACT**14** in the flowchart in FIG. **4**, or can also perform ACT**13** and ACT**14** in parallel to each other. As described above, as long as no conflict with preceding or following processing occurs, it is possible to modify the order of the processing or to perform a plurality of types of processing in parallel to each other.

(207) In addition, although some embodiments of the present disclosure are described, these embodiments are illustrative only, and it is not intended to limit the scope of the present disclosure. These novel embodiments can be implemented with other various aspects, and a variety of omissions, replacements, and modifications can be made within the scope or the spirit of the present disclosure. These embodiments and modifications thereof are included in the scope of the present disclosure, and at the same time, included in the present disclosure set forth in the appended claims, and the equivalents thereof.

Claims

1. A control device configured to control annunciation members disposed so as to correspond respectively to a plurality of behavior areas where a plurality of behaviors included in a series of behaviors to be executed by a shopper is executed, the control device comprising: a behavior recognition component configured to recognize a behavior of the plurality of behaviors executed by the shopper; a determination component configured to determine a subsequent behavior of the plurality of behaviors as a behavior for the shopper to subsequently perform based on the behavior recognized by the behavior recognition component; a subsequent behavior annunciation component configured to make an annunciation member disposed so as to correspond to a behavior area where the subsequent behavior is to be executed operate in a first annunciation configuration based on the subsequent behavior determined by the determination component; a correct behavior annunciation component configured to make the annunciation member, which operates in the first annunciation configuration, operate in a second annunciation configuration different from the first annunciation configuration when the behavior recognition component recognizes that the shopper executes the subsequent behavior; and a warning component configured to make the annunciation member, which is disposed so as to correspond to the behavior area where a different behavior from the subsequent behavior is executed, operate in a warning configuration different from the first annunciation configuration and the second annunciation configuration when the behavior recognition component recognizes that the shopper executes the different behavior, wherein the plurality of behaviors include a scanning behavior of scanning a code symbol attached to a purchased article with a scanner and a behavior to place a scanned article on a predetermined area.
2. The control device according to claim 1, wherein the series of behaviors to be executed by the shopper include a plurality of behaviors to be performed by the shopper when executing

registration of the purchased article, the control device further includes a storage component configured to store an order of the plurality of behaviors to be performed when executing the registration of the purchased article, and the determination component is further configured to determine the subsequent behavior based on the order of the behavior stored in the storage component.

3. The control device according to claim 2, wherein the plurality of behaviors includes at least the scanning behavior of scanning the code symbol attached to the purchased article with the scanner, or an imaging behavior of making an imaging component image an appearance of the purchased article, and the behavior recognition component is further configured to recognize that the shopper executes the scanning behavior or the imaging behavior based on an output of the scanner or the imaging component.

4. The control device according to claim 1, wherein the annunciation members include a visual annunciation member, and the first annunciation configuration by the subsequent behavior annunciation component, the second annunciation configuration by the correct behavior annunciation component, and the warning configuration by the warning component are different in visual annunciation configuration of the visual annunciation member from each other.

5. The control device according to claim 4, wherein the visual annunciation member includes a light emitting member, and the first annunciation configuration by the subsequent behavior annunciation component, the second annunciation configuration by the correct behavior annunciation component, and the warning configuration by the warning component unit are different in at least one of a color of light emitted by the light emitting member and a lighting pattern of the light emitting member from each other.

6. The control device according to claim 1, wherein the control device is comprised in a self-service POS terminal.

7. The control device according to claim 1, wherein the behavior area comprises a basket table area.

8. A behavior guide system, comprising: annunciation members disposed so as to correspond respectively to a plurality of behavior areas where a plurality of behaviors included in a series of behaviors to be executed by a shopper is executed, wherein the plurality of behaviors include a scanning behavior of scanning a code symbol attached to a purchased article with a scanner and a behavior to place a scanned article on a predetermined area; and a control device configured to control the annunciation members, wherein the control device includes a behavior recognition component configured to recognize a behavior of the plurality of behaviors executed by the shopper, a determination component configured to determine a subsequent behavior of the plurality of behaviors as a behavior for the shopper to subsequently perform based on the behavior recognized by the behavior recognition component, a subsequent behavior annunciation component configured to make an annunciation member disposed so as to correspond to a behavior area where the subsequent behavior is to be executed operate in a first annunciation configuration based on the subsequent behavior determined by the determination component, a correct behavior annunciation component configured to make the annunciation member, which operates in the first annunciation configuration, operate in a second annunciation configuration different from the first annunciation configuration when the behavior recognition component recognizes that the shopper executes the subsequent behavior, and a warning component configured to make the annunciation member, which is disposed so as to correspond to the behavior area where a different behavior from the subsequent behavior is executed, operate in a warning configuration different from the first annunciation configuration and the second annunciation configuration when the behavior recognition component recognizes that the shopper executes the different behavior.

9. The behavior guide system according to claim 8, wherein the series of behaviors to be executed by the shopper include a plurality of behaviors to be performed by the shopper when executing registration of the purchased article, the control device further includes a storage component

configured to store an order of the plurality of behaviors to be performed when executing the registration of the purchased article, and the determination component is further configured to determine the subsequent behavior based on the order of the behavior stored in the storage component.

10. The behavior guide system according to claim 9, wherein the plurality of behaviors includes at least the scanning behavior of scanning the code symbol attached to the purchased article with the scanner, or an imaging behavior of making an imaging component image an appearance of the purchased article, and the behavior recognition component is further configured to recognize that the shopper executes the scanning behavior or the imaging behavior based on an output of the scanner or the imaging component.

11. The behavior guide system according to claim 8, wherein the annunciation members include a visual annunciation member, and the first annunciation configuration by the subsequent behavior annunciation component, the second annunciation configuration by the correct behavior annunciation component, and the warning configuration by the warning component are different in visual annunciation configuration of the visual annunciation member from each other.

12. The behavior guide system according to claim 11, wherein the visual annunciation member includes a light emitting member, and the first annunciation configuration by the subsequent behavior annunciation component, the second annunciation configuration by the correct behavior annunciation component, and the warning configuration by the warning component unit are different in at least one of a color of light emitted by the light emitting member and a lighting pattern of the light emitting member from each other.

13. The behavior guide system according to claim 8, wherein the control device is comprised in a self-service POS terminal.

14. The behavior guide system according to claim 8, wherein the behavior area comprises a basket table area.

15. A method for controlling a behavior guide system including a processor, a memory, and annunciation members disposed so as to correspond respectively to a plurality of behavior areas where a plurality of behaviors included in a series of behaviors to be executed by a shopper is executed, the method comprising: recognizing a behavior of the plurality of behaviors executed by the shopper; determining a subsequent behavior of the plurality of behaviors as a behavior for the shopper to subsequently perform based on the behavior recognized in the recognizing the behavior executed by the shopper and an order of the plurality of behaviors stored in the memory; making an annunciation member disposed so as to correspond to a behavior area where the subsequent behavior is to be executed operate in a first annunciation configuration based on the subsequent behavior determined in the determining the subsequent behavior as the behavior for the shopper to subsequently perform; make the annunciation member, which operates in the first annunciation configuration, operate in a second annunciation configuration different from the first annunciation configuration when recognized in the recognizing the behavior executed by the shopper that the shopper executes the subsequent behavior; and making the annunciation member, which is disposed so as to correspond to the behavior area where a different behavior from the subsequent behavior is executed, operate in a warning configuration different from the first annunciation configuration and the second annunciation configuration when recognized in the recognizing the behavior executed by the shopper that the shopper executes the different behavior, wherein the plurality of behaviors include a scanning behavior of scanning a code symbol attached to a purchased article with a scanner and a behavior to place a scanned article on a predetermined area.

16. The method according to claim 15, wherein the series of behaviors to be executed by the shopper include a plurality of behaviors to be performed by the shopper when executing registration of the purchased article, the method further comprising: storing the order of the plurality of behaviors to be performed when executing the registration of the purchased article, and determining the subsequent behavior based on the order of the behavior stored.

17. The method according to claim 16, wherein the plurality of behaviors includes at least the scanning behavior of scanning the code symbol attached to the purchased article with the scanner, or an imaging behavior of making an imaging component image an appearance of the purchased article, the method further comprising: recognizing that the shopper executes the scanning behavior or the imaging behavior based on an output of the scanner or the imaging.
 18. The method according to claim 15, wherein the annunciation members include a visual annunciation member, and the first annunciation configuration, the second annunciation configuration by the correct behavior annunciation, and the warning configuration are different in visual annunciation configuration of the visual annunciation member from each other.
 19. The method according to claim 18, wherein the visual annunciation member includes a light emitting member, and the first annunciation configuration, the second annunciation configuration, and the warning configuration are different in at least one of a color of light emitted by the light emitting member and a lighting pattern of the light emitting member from each other.
 20. The method according to claim 15, wherein the behavior area comprises a basket table area.
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